

SECTION A

1. Five Unique Properties That Make Water “The Fluid of Life”

1. **Cohesion and Adhesion:** Water molecules stick to each other (cohesion) and to surfaces (adhesion), enabling processes like water transport in plants (capillary action).
 2. **High Heat Capacity:** Water absorbs and releases heat slowly, stabilizing temperatures in organisms and environments.
 3. **Universal Solvent:** Water dissolves many substances, allowing for chemical reactions in cells and transport of nutrients and waste.
 4. **Ice Floats:** Water is less dense as a solid than as a liquid, allowing ice to float and insulate aquatic life in cold climates.
 5. **High Surface Tension:** This property enables water to form droplets and support small organisms on its surface.
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2. Simplest Forms of Life and Whether Viruses are Alive

- **Simplest Forms of Life:** Prokaryotes (e.g., bacteria and archaea) are unicellular organisms with no nucleus, performing all life functions within one cell.
 - **Are Viruses Alive?:** Viruses are not considered fully alive because:
 - They lack cellular structure and metabolism.
 - They cannot reproduce without a host.However, they exhibit characteristics of life (e.g., genetic material) inside a host, blurring the definition.
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3. Pangenesis vs. Germplasm Theory

- **Pangenesis (Charles Darwin):** Suggests that all body cells release "gemmules" carrying traits to reproductive cells, influencing heredity.
 - **Germplasm Theory (August Weismann):** States that hereditary information is passed only through germ cells (sperm and egg), while somatic cells (body cells) do not influence heredity.
The germplasm theory forms the basis of modern genetics.
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4. Definitions

a) **Coacervate:** Droplet-like structures formed from macromolecules in water, considered precursors to cell life.

- b) **Homunculus:** A historical theory that sperm or eggs contained a tiny preformed human.
 - c) **Acrosome:** A cap-like structure on sperm that releases enzymes to penetrate the egg during fertilization.
 - d) **Placebo:** An inactive treatment used in experiments to test the effectiveness of a drug.
 - e) **Phylogenesis:** The evolutionary development and history of a species.
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5. Five Statements of Modern Cell Theory

1. All living organisms are made up of one or more cells.
 2. The cell is the basic unit of life.
 3. All cells arise from pre-existing cells.
 4. Cells carry genetic information (DNA), passed during cell division.
 5. All energy flow (metabolism) occurs within cells.
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6. Plastids

Plastids are double-membrane organelles found in plants and algae. Types include:

- **Chloroplasts:** Perform photosynthesis.
 - **Chromoplasts:** Store pigments (e.g., in fruits and flowers).
 - **Leucoplasts:** Store nutrients like starch, oils, or proteins.
- Plastids play key roles in energy production and storage.
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7. Cell Description

- a) **Types:** Prokaryotic (no nucleus, e.g., bacteria) and Eukaryotic (with nucleus, e.g., plants, animals).
 - b) **Sizes:** Prokaryotes (1–10 μm), Eukaryotes (10–100 μm).
 - c) **Structures:** Both have a plasma membrane, cytoplasm, and ribosomes. Eukaryotes also have organelles like the nucleus, mitochondria, and Golgi apparatus.
 - d) **Numbers:** Unicellular (e.g., bacteria) or Multicellular (e.g., humans).
 - e) **Shapes:** Vary from spherical (cocci) to rod-shaped (bacilli), spiral (spirilla), or irregular.
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8. Why the Golgi Apparatus is Referred to as the Most Changing Organelle

The **Golgi Apparatus** constantly changes because it processes, modifies, and transports proteins and lipids:

- Proteins and lipids from the Endoplasmic Reticulum enter the Golgi as vesicles.
- The Golgi modifies these molecules (e.g., adding sugars) and packages them into new vesicles.

- It dynamically assembles and disassembles as vesicles fuse or leave its structure.

Illustration Example

The Golgi appears as a stack of flattened sacs (cisternae) with incoming vesicles on the "cis" face and outgoing vesicles on the "trans" face. Here's a step-by-step diagram idea:

1. Vesicles fuse with the **cis-Golgi**.
2. Molecules are modified in the Golgi stacks.
3. Finished products exit from the **trans-Golgi** to their destinations (e.g., cell membrane or lysosomes).

SECTION B

1. Discuss the Steps and Process of the Scientific Method

1. **Observation:** Identify a problem or question based on natural phenomena.
 2. **Research:** Gather background information and existing knowledge.
 3. **Ask a Question:** Frame a clear, concise question to investigate.
 4. **Form a Hypothesis:** Develop a testable prediction to explain the observation.
 5. **Experiment Design:** Create a controlled experiment to test the hypothesis.
 6. **Variable Identification:** Define independent, dependent, and controlled variables.
 7. **Conduct Experiment:** Perform the experiment and collect data.
 8. **Record Data:** Use qualitative and quantitative methods to record observations.
 9. **Analyze Data:** Use statistical tools to interpret findings and identify trends.
 10. **Draw Conclusions:** Determine if the hypothesis was supported or rejected.
 11. **Refine Hypothesis:** Revise the hypothesis if necessary and repeat the process.
 12. **Develop a Theory:** When multiple tests confirm a hypothesis, it may evolve into a theory.
 13. **Peer Review:** Share findings with the scientific community for evaluation.
 14. **Publish Results:** Write a detailed report or paper for publication.
 15. **Replication:** Encourage other scientists to replicate the study for validation.
 16. **Controlled Testing:** Ensure reproducibility by testing under controlled conditions.
 17. **Reassessment:** Adjust the model or hypothesis based on new evidence.
 18. **Ethical Considerations:** Ensure that research complies with ethical standards.
 19. **Problem-Solving:** Use findings to address real-world challenges.
 20. **Technological Application:** Apply findings to develop new tools or methods.
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2. Evaluate Various Theories on the Origin of Life on Earth

1. **Abiogenesis:** Life arose spontaneously from non-living matter through chemical evolution.
2. **Miller-Urey Experiment:** Simulated early Earth conditions, forming amino acids.
3. **Panspermia Theory:** Life or its precursors were delivered to Earth via meteorites or comets.

4. **Hydrothermal Vent Hypothesis:** Life began in deep-sea hydrothermal vents with stable, mineral-rich environments.
 5. **Primordial Soup Hypothesis:** Earth's early oceans had organic molecules, forming life under energy sources like lightning or UV radiation.
 6. **RNA World Hypothesis:** RNA was the first self-replicating molecule, leading to life.
 7. **Iron-Sulfur World Hypothesis:** Suggests life originated on mineral surfaces, catalyzing biochemical reactions.
 8. **Clay Hypothesis:** Clay surfaces acted as catalysts for organic molecule formation and polymerization.
 9. **Electric Spark Hypothesis:** Lightning provided the energy to create organic molecules from atmospheric gases.
 10. **Deep-Sea Alkaline Hydrothermal Theory:** Alkaline hydrothermal vents created a chemical gradient, fostering molecule formation.
 11. **Frozen Ocean Theory:** Ice-covered oceans protected early molecules from UV radiation, aiding polymerization.
 12. **Thermal Pool Hypothesis:** Pools of volcanic activity concentrated organic molecules for reactions.
 13. **Directed Panspermia:** Life was intentionally seeded on Earth by extraterrestrial civilizations.
 14. **Coacervates:** Droplets of organic molecules formed precursors to cells.
 15. **Endosymbiotic Theory:** Early cells developed by engulfing simpler cells, leading to eukaryotes.
 16. **Creationism:** Life was created by a divine being (faith-based explanation).
 17. **Biopoiesis:** Gradual chemical processes led to the transition from molecules to living organisms.
 18. **Entropy and Life:** The origin of life is linked to systems reducing entropy locally while increasing universal entropy.
 19. **Criticism of Abiogenesis:** Critics argue abiogenesis cannot fully explain self-replicating molecules.
 20. **Current Consensus:** Abiogenesis with support from hydrothermal vents and RNA World Hypothesis is widely accepted in science.
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3. Write an Essay on Genetic Materials

Introduction:

Genetic material is the foundation of inheritance, controlling the transfer of traits and life functions. The primary genetic materials are **DNA (deoxyribonucleic acid)** and **RNA (ribonucleic acid)**.

Structure of Genetic Material:

1. **DNA:**
 - Double-stranded helix made of nucleotides (adenine, thymine, cytosine, guanine).
 - Sugar-phosphate backbone with complementary base pairing (A-T, C-G).
2. **RNA:**
 - Single-stranded molecule with uracil replacing thymine.

- Types: mRNA (messenger RNA), tRNA (transfer RNA), rRNA (ribosomal RNA).

Functions of Genetic Material:

1. **Storage of Information:** DNA stores genetic instructions for cellular functions.
2. **Transmission of Traits:** DNA passes genetic traits from parents to offspring.
3. **Expression of Genes:** Information is transcribed into RNA and translated into proteins.
4. **Mutation:** Changes in DNA sequence generate genetic variation and evolution.
5. **Replication:** DNA replicates during cell division to ensure genetic continuity.

DNA vs. RNA:

1. DNA is stable, stores long-term genetic information, and is found in the nucleus.
2. RNA is less stable, acts as an intermediary, and assists in protein synthesis.

Importance of Genetic Material:

1. **Cellular Function:** Directs processes like metabolism, growth, and development.
2. **Medical Applications:** Understanding DNA aids in genetic engineering, gene therapy, and disease treatment.
3. **Evolution:** Genetic changes drive adaptation and speciation.

Modern Advances in Genetics:

1. **CRISPR-Cas9:** Revolutionized gene editing for precise modifications.
2. **Human Genome Project:** Mapped all human genes, advancing personalized medicine.
3. **Gene Therapy:** Treats genetic disorders by inserting functional genes.

Conclusion:

Genetic material is the key to understanding life and its complexity. Advances in genetic research continue to transform medicine, agriculture, and biotechnology, offering solutions to global challenges.

SECTION A

1. Functions, Structure, and Distribution of Centrioles

- **Structure:** Centrioles are cylindrical structures composed of microtubules arranged in a 9+0 pattern (nine triplets of microtubules without a central pair). They are part of the centrosome, which is located near the nucleus in animal cells.
- **Functions:**

1. **Cell Division:** Centrioles play a critical role in organizing the mitotic spindle during mitosis and meiosis, ensuring proper chromosome separation.
 2. **Formation of Cilia and Flagella:** They help in the formation of basal bodies, which are the foundations for cilia and flagella in eukaryotic cells.
 - **Distribution:** Centrioles are found in animal cells and some lower plants. They are typically located near the nucleus within the centrosome region, and they replicate during cell division.
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2. Differentiate Prokaryotes and Eukaryotes on the Following:

- **Genetic Material:**
 - **Prokaryotes:** Have a single, circular DNA molecule located in the nucleoid region, without a membrane-bound nucleus.
 - **Eukaryotes:** Have linear DNA enclosed in a membrane-bound nucleus.
 - **Respiration:**
 - **Prokaryotes:** Cellular respiration occurs across the cell membrane (e.g., in the cytoplasm or cell membrane).
 - **Eukaryotes:** Respiration mainly occurs in mitochondria (a membrane-bound organelle).
 - **Photosynthesis:**
 - **Prokaryotes:** Some prokaryotes (like cyanobacteria) perform photosynthesis using their cell membrane.
 - **Eukaryotes:** Plants and algae have chloroplasts, specialized organelles, for photosynthesis.
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3. Explain Why the Genetic Code is Considered:

- **Nearly Universal:** The genetic code is nearly universal because it is shared by nearly all organisms, from bacteria to humans, making it fundamental for protein synthesis across species.
 - **Triplet:** The genetic code is triplet because three nucleotides (codons) in DNA or RNA specify a single amino acid in a protein.
 - **Non-overlapping:** The genetic code is non-overlapping because each nucleotide is part of only one codon; there is no sharing of nucleotides between adjacent codons.
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4. Reasons Why DNA is Considered the Primary Genetic Substance

1. **Stability:** DNA is chemically stable, allowing it to store genetic information for long periods.
2. **Replication:** DNA can replicate accurately during cell division, ensuring that genetic information is passed on to daughter cells.
3. **Mutation:** DNA undergoes mutations that can lead to genetic variation and evolution.

4. **Protein Encoding:** DNA contains the instructions for making proteins, which are essential for cellular structure and function.
 5. **Inheritance:** DNA is passed from parents to offspring, carrying the traits that define organisms.
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5. State and Explain the Cell Theory

- **Cell Theory** states that:
 1. **All living organisms are made of cells:** Cells are the basic building blocks of life, forming the structure of all organisms.
 2. **The cell is the basic unit of structure and function:** All functions of life, such as metabolism and reproduction, occur within cells.
 3. **All cells arise from pre-existing cells:** New cells are formed by the division of existing cells, ensuring continuity of life.
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6. Structure and Distribution of Lysosomes

- **Structure:** Lysosomes are membrane-bound organelles containing digestive enzymes that break down waste materials and cellular debris.
 - **Distribution:** They are primarily found in animal cells and are scattered throughout the cytoplasm. They can fuse with vacuoles or other cellular structures to degrade their contents.
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7. Explain How Substances Go Into and Out of Cells in Bulk

- **Endocytosis:** The process by which cells engulf large particles or liquids using their plasma membrane, forming vesicles to bring substances into the cell. This includes:
 1. **Phagocytosis:** Engulfment of solid particles.
 2. **Pinocytosis:** Engulfment of liquids.
 - **Exocytosis:** The process where cells expel large molecules, such as waste or secretory proteins, by vesicles fusing with the plasma membrane.
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8. Explain the Main Differences in Structure and Functions of DNA and RNA Genetic Materials

- **Structure:**
 - **DNA:** Double-stranded helix, contains deoxyribose sugar, and uses thymine (T) as one of the nitrogenous bases.
 - **RNA:** Single-stranded, contains ribose sugar, and uses uracil (U) instead of thymine.
- **Functions:**

- **DNA:** Stores genetic information and serves as a template for RNA synthesis.
- **RNA:** Involved in protein synthesis, including messenger RNA (mRNA) that carries genetic instructions from DNA to ribosomes, transfer RNA (tRNA) that helps assemble proteins, and ribosomal RNA (rRNA) that is part of the ribosome structure.

SECTION B

1. Write an essay on the modern theory of the origin of cells and life.

The modern theory of the origin of cells and life explains the transition from non-living molecules to living organisms through a series of chemical and biological processes. The key aspects are as follows:

1. **Abiogenesis Hypothesis:** Life originated from non-living matter through a natural process. Early Earth's conditions allowed simple organic molecules to form.
2. **Primordial Soup:** Proposed by Alexander Oparin and J.B.S. Haldane, Earth's oceans were rich in organic compounds that acted as precursors to life.
3. **Miller-Urey Experiment:** Stanley Miller and Harold Urey demonstrated the formation of amino acids (basic building blocks of life) by simulating early Earth's conditions.
4. **Formation of Polymers:** Organic molecules like amino acids and nucleotides polymerized to form proteins and nucleic acids.
5. **Protocells:** Lipid molecules formed vesicles, creating structures similar to cell membranes, which encapsulated organic material.
6. **RNA World Hypothesis:** RNA acted as the first genetic material and catalyst due to its dual role in information storage and enzymatic activity.
7. **Self-replicating Molecules:** Molecules capable of replication emerged, leading to heredity and evolution.
8. **Metabolic Pathways:** Primitive metabolic pathways evolved to harness energy for survival and reproduction.
9. **Transition to DNA and Proteins:** DNA replaced RNA as a more stable genetic material, while proteins became the primary enzymes.
10. **Evolution of Cells:** Protocells evolved into prokaryotic cells, with structures to support metabolism and replication.
11. **Endosymbiotic Theory:** Eukaryotic cells formed through symbiosis between early prokaryotic cells, where mitochondria and chloroplasts originated from engulfed bacteria.
12. **Role of Hydrothermal Vents:** Deep-sea hydrothermal vents provided energy and chemical compounds, creating a conducive environment for life to emerge.
13. **Panspermia Hypothesis:** Some suggest life's building blocks came from extraterrestrial sources, though this is not universally accepted.
14. **Natural Selection:** Variations among early cells allowed for selection and evolution, leading to the diversity of life forms.
15. **Modern Evidence:** Fossilized microbial life and molecular biology techniques support these theories, confirming the gradual development of cells and life.

2. Describe the structure, distribution, and functions of cell membranes

Cell membranes, also known as plasma membranes, are vital for maintaining the structure and function of cells.

Structure of Cell Membranes

1. **Lipid Bilayer:** Composed of a double layer of phospholipids with hydrophilic (water-attracting) heads and hydrophobic (water-repelling) tails.
2. **Proteins:** Integral and peripheral proteins are embedded within or attached to the bilayer, performing structural and functional roles.
3. **Carbohydrates:** Glycoproteins and glycolipids on the surface are involved in cell recognition and signaling.
4. **Cholesterol:** Embedded within the membrane, it maintains fluidity and stability, especially in animal cells.
5. **Fluid Mosaic Model:** Describes the dynamic nature of the membrane where lipids and proteins can move laterally within the layer.

Distribution of Cell Membranes

1. **Plasma Membrane:** Encloses the cell, separating it from the external environment.
2. **Organelle Membranes:** Found in eukaryotic organelles like the nucleus, mitochondria, and endoplasmic reticulum, compartmentalizing cell functions.
3. **Outer Membranes:** Bacteria and some organelles, like chloroplasts, have double membranes for protection and function.

Functions of Cell Membranes

1. **Selective Permeability:** Regulates the entry and exit of substances, maintaining homeostasis.
2. **Protection:** Acts as a barrier against harmful substances.
3. **Communication:** Contains receptors that detect and respond to chemical signals.
4. **Transport:** Facilitates passive (diffusion, osmosis) and active transport (endocytosis, exocytosis) of molecules.
5. **Enzyme Activity:** Provides a site for enzymatic reactions, such as ATP synthesis in mitochondria.
6. **Structural Support:** Anchors the cytoskeleton, maintaining cell shape.
7. **Cell-Cell Interaction:** Involved in adhesion, recognition, and communication between cells.
8. **Energy Production:** In prokaryotes, the membrane assists in processes like respiration and photosynthesis.
9. **Endocytosis and Exocytosis:** Facilitates the uptake of materials and secretion of waste.
10. **Antigen Presentation:** Plays a role in immune responses by displaying antigens.
11. **Signal Transduction:** Converts extracellular signals into intracellular actions.
12. **Water Balance:** Controls osmosis to prevent cell lysis or dehydration.
13. **Lipid and Protein Synthesis:** Serves as a scaffold for synthesizing membrane components.
14. **Storage:** Stores specific enzymes and ions necessary for cell activity.

15. **Evolutionary Role:** Membranes contributed to the origin of compartmentalization, critical for eukaryotic evolution.
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3. Describe the main characteristics of living things

Living things exhibit specific characteristics that distinguish them from non-living entities:

1. **Cellular Organization:** All living organisms are made up of cells, the basic unit of life. Organisms can be unicellular (e.g., bacteria) or multicellular (e.g., humans).
2. **Growth and Development:** Living things grow by increasing in size or cell number and undergo developmental changes throughout their lifecycle.
3. **Metabolism:** Living organisms perform chemical reactions to obtain and utilize energy, including catabolism (breaking down molecules) and anabolism (synthesizing molecules).
4. **Reproduction:** All living things reproduce to ensure the continuity of their species, either sexually or asexually.
5. **Response to Stimuli:** Organisms react to environmental changes such as light, temperature, or danger (e.g., plants bending towards light).
6. **Adaptation:** Through evolution, living organisms adapt to their environment to enhance survival and reproduction.
7. **Homeostasis:** The ability to maintain a stable internal environment despite external fluctuations (e.g., regulating body temperature).
8. **Heredity:** Organisms pass genetic material (DNA or RNA) to their offspring, ensuring continuity and variation.
9. **Movement:** Organisms exhibit movement, either of the whole body (animals) or internal structures (cytoplasmic streaming in plants).
10. **Excretion:** Living things eliminate metabolic waste products to maintain cellular function.
11. **Respiration:** Organisms convert glucose into energy (ATP) through cellular respiration.
12. **Complex Organization:** Living things have complex structures, including organelles, tissues, and organ systems in higher organisms.
13. **Ecological Interaction:** Living organisms interact with their environment and other organisms (e.g., symbiosis, predation).
14. **Sensitivity:** Ability to detect and respond to internal or external environmental conditions.
15. **Finite Lifespan:** All living things have a defined lifecycle, including birth, growth, reproduction, and death.